

Dynamic Split Body: Changing Body Perception and Self-Location by Manipulating Half-Body Position

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Figure 1: Varieties of split body: (a) narrow split, (b) wide split, (c) front-back split, (d) left-right swap

ABSTRACT

Some studies have been conducted to induce the sense of being in two locations by inducing body ownership in two bodies. However, the feeling of being in two locations was weak. In our previous study, we induced body ownership in a split avatar in advance to induce the sensation of being at two locations. As a result, although participants' self-location extended to the right, the split avatar was perceived as a single body and they did not feel as if they were in two locations. Our demonstration expand our previous work by allowing participants to dynamically manipulate the position of the split body. We attempt to present an experience in which the split body is perceived as one body even though it is split, or the participants feel as if they are in two locations as two independent bodies. We also changed tracking system from OptiTrack to Lighthouse.

CCS CONCEPTS

• Human-centered computing → Virtual reality.

KEYWORDS

embodiment, body ownership, self-location, split body

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1 INTRODUCTION

Visual-tactile synchronization [Lenggenhager et al. 2007] and visual-motor synchronization [Gonzalez-Franco et al. 2010] make a non-native body feel as if it were one's own body (Illusory body ownership). Then, when body ownership is generated for a body presented in a position different from the observer's body, the perceived self-location drifts to the presented body position [Lenggenhager et al. 2007]. In the past, the method to manipulate distant objects by shifting the position of the virtual hand has also been proposed (Go-Go interaction [Poupyrev et al. 1996]).

Illusory body ownership also occurs for multiple bodies. In Guterstam et al. [Guterstam et al. 2020], two mannequins were presented through a head-mounted display (HMD), and simultaneous stroking of a participant's body and the two mannequins produced body ownership in two mannequins. In addition, by switching the view-points of the two mannequins in time, participants feel as if they are in two different locations. In Aymerich-Franch et al [Aymerich-Franch et al. 2016], a participant operated a teleoperated robot. The participant observed their own body from the robot's perspective, making both the robot and the participant's body feel like their own bodies and causing them to feel as if they were in both the robot and the participant's body locations.

Thus, although self-location is attempted to be extended by using multiple bodies, the body ownership in multiple bodies and the sense of being at two different locations are weak. It is thought that because of the one-to-two mapping of the bodies between the observer's body and the presented body.

In our previous study [Kondo and Sugimoto 2022], we investigated whether the self-location could be extended to two spots by splitting a full-body avatar, which corresponded one body to one participant's body, to the left or right, while maintaining the same correspondence as a normal body. The results showed that the self-location extended to the right side when the body was split. However, even when the body was split, it was perceived as a single

body spread horizontally, suggesting that the participant felt as if they are in one location. In addition, a split avatar was presented beforehand, and the participant was not allowed to split the avatar and move it. Depending on the arrangement of the split avatars, they could be perceived as two independent bodies, which could be perceived as if they were in two locations.

In this demonstration, we expanded our previous work and developed an environment that allows participants to manipulate the position of the split body. As a result, various split body arrangements are possible, participants may experience both the sensation of being in one location with one body and the sense of being in two locations with two bodies.

2 THE DEMONSTRATION

2.1 Materials

A participant observes a virtual environment through an HTC Vive Pro HMD (Fig. 2). In addition, the participant holds Valve index controllers in their left and right hands and wears Vive trackers on their left and right feet and waist to perform full body tracking using Virtual Motion Capture¹. Their motions are reflected to the split avatar in Unity using Easy Virtual Motion Capture For Unity².

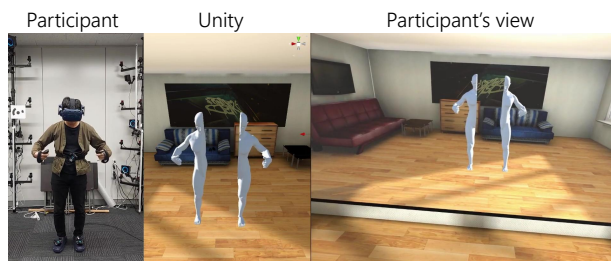


Figure 2: The demonstration of dynamic split body

2.2 Procedure

The participant wears the device, calibrates the avatar, and then observes the VR space, where a pre-split avatar is presented at the same position as the participant's body, which is observed from a first-person perspective. A mirror is placed in front of the participant. When the participant moves their body, the avatar moves in synchronization with their movement, thereby making the avatar feel like their own body. The participant moves the controller horizontally while pressing the trigger on the controller, and half of the avatar moves horizontally corresponding to the amount of the controller's movement. The participant's viewpoint does not change at the time. The right controller is used to move the right half of the body, and the left controller is used to move the left half of the body. The half-body position can be reset by pressing the button on the right controller. Thus, the participant can freely manipulate the positions of the left and right halves of the body by operating the controllers. Since the avatar moves in synchronization with the participant's movement after splitting, body ownership is maintained, and the participant can experience

the sensation that their body has been split and as if they were in the position of the split body.

2.3 Varieties of split body

Since the half-body positions can be freely manipulated, various split bodies can be experienced. For example, when the left and right halves are extended horizontally, if the distance between the half-bodies is close, as in the previous study, participants will be perceived as one body, including the empty space between the half-bodies (Fig. 1 a). If the distance between the two bodies is too far, participants will feel as if they have two separate bodies (Fig. 1 b). When participants move the split avatar back and front (Fig. 1 c) or swaps left and right (Fig. 1 d), they may feel like they have two bodies. In this way, we attempt to provide an experience in which the perception of the body changes dynamically depending on the position of the two bodies.

The perception of a split body as a single body could be related to the case that when the hands and feet-only body has a layout similar to that of the human body structure, the body between them is complemented [Kondo et al. 2020]. When this layout is disrupted, the perception of the body as one coherent body is lost, and the split body may be perceived as two independent bodies.

3 CONCLUSION

In this demonstration, a participant's motion is reflected in the avatar in real-time, and by splitting the avatar, the participant will experience their body splitting while maintaining body ownership and the experience of being located in two different locations. In addition, since the position of the split half-body can be freely manipulated by the participant, the perception of the body would dynamically change depending on the position of the half-body, such as one wide body or two independent bodies. This study would allow humans to flexibly change their bodies according to tasks such as one large body or two half-bodies.

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¹<https://vmc.info/>

²<https://gpsnmeajp.github.io/EasyVirtualMotionCaptureForUnity-documents/>